

COMP 484 HW10 - Chrome DevTools Report

1) What I Implemented

I copied my Project 2 Giga Pet into a new HW10 project and added Chrome DevTools practice features. I built buttons that generate console logs (info, warning, error, table, group, and custom), browser issues (404, TypeError, and violation), and filter-friendly messages (level, text, regex, source, and user tags). I also added a buggy addition example for debugging and a fixed version using Number() so I could demonstrate breakpoint debugging, variable inspection, and applying a fix. The bug is adding two values whose type is string ("5" + "1" = "51"), instead of intended mathematical addition.

Giga Pet DevTools

My Project 2 copy with some Chrome DevTools examples added.



Chrome DevTools Practice

Open Chrome DevTools and click the buttons to view output.

Message Logging

Log InfoLog WarningLog ErrorLog TableLog GroupLog Custom

ElementsConsoleSourcesNetworkPerformanceMemoryApplicationSecurityLighthouseRecorder

topFilter

PET-INFO: DevTools homework startedscript.js:12

PET-INFO: Current pet state {name: 'Koa', weight: 70, happiness: 7, energy: 7}script.js:56

PET-INFO: Koa is ready.script.js:225

PET-INFO: Info-level message: happiness is 7script.js:226

PET-WARN: Koa's energy is getting checked.script.js:230

PET-WARN: Trace for warning examplescript.js:231

logWarningExample @ script.js:231

dispatch @ jquery-3.7.1.min.js:2

ce.event.add.v.handle @ jquery-3.7.1.min.js:2

PET-ERROR: Example error message for the console.script.js:235

script.js:246

(index)	action	happinessChange	energyChange	weightChange
0	'Treat'	'+1'	'0'	'+1'
1	'Play'	'+1'	'-1'	'-1'
2	'Exercise'	'-1'	'-1'	'-1'
3	'Nap'	'+1'	'+3'	'0'

Array(4)

PET-INFO: Koa checklistscript.js:250

PET-INFO: Food checkedscript.js:251

PET-INFO: Played outsidescript.js:252

PET-INFO: Nap time laterscript.js:253

PET-INFO: Need screenshotsscript.js:254

FILTER-DEMO: Custom styled console message.script.js:259

2) Message Logging

Implemented examples:

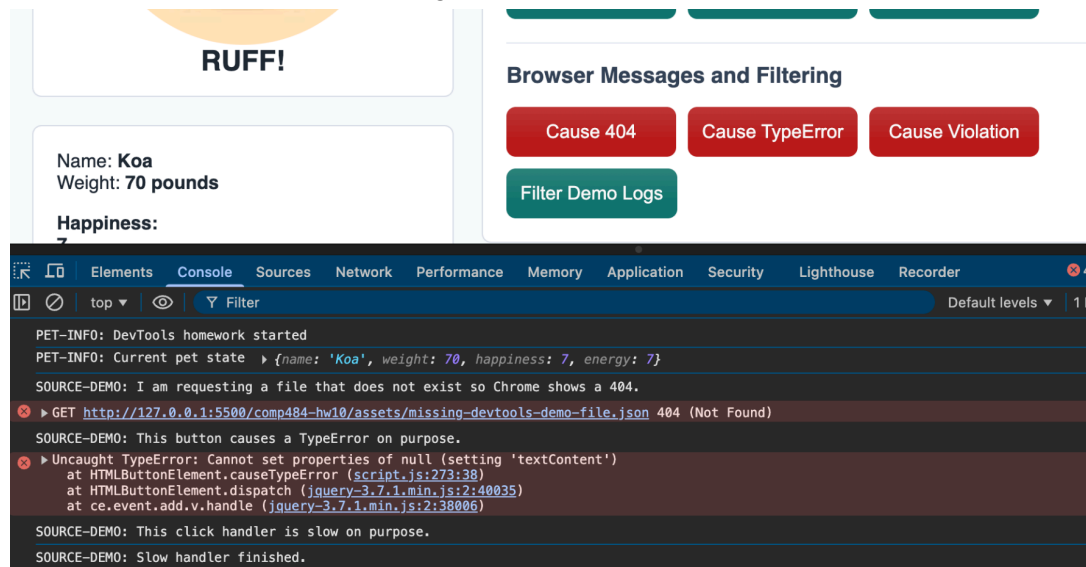
Buttons → functions:

- Log Info → ``logInfoExample()`` (button: Log Info)
- Log Warning → ``logWarningExample()`` (button: Log Warning)
- Log Error → ``logErrorExample()`` (button: Log Error)
- Log Table → ``logTableExample()`` (button: Log Table)
- Log Group → ``logGroupExample()`` (button: Log Group)
- Log Custom → ``logCustomExample()`` (button: Log Custom)

3) View Messages Logged by the Browser

Implemented examples:

- Cause 404: clicking the Cause 404 button requests a missing asset (``fetch("assets/missing-devtools-demo-file.json")``) and shows a 404 in Network/Console.
- Cause TypeError: clicking the TypeError button accesses a property on ``null``, producing a runtime `TypeError`.
- Cause Violation: clicking the Violation button runs a deliberate slow loop (blocking work) so DevTools reports a long task/violation.

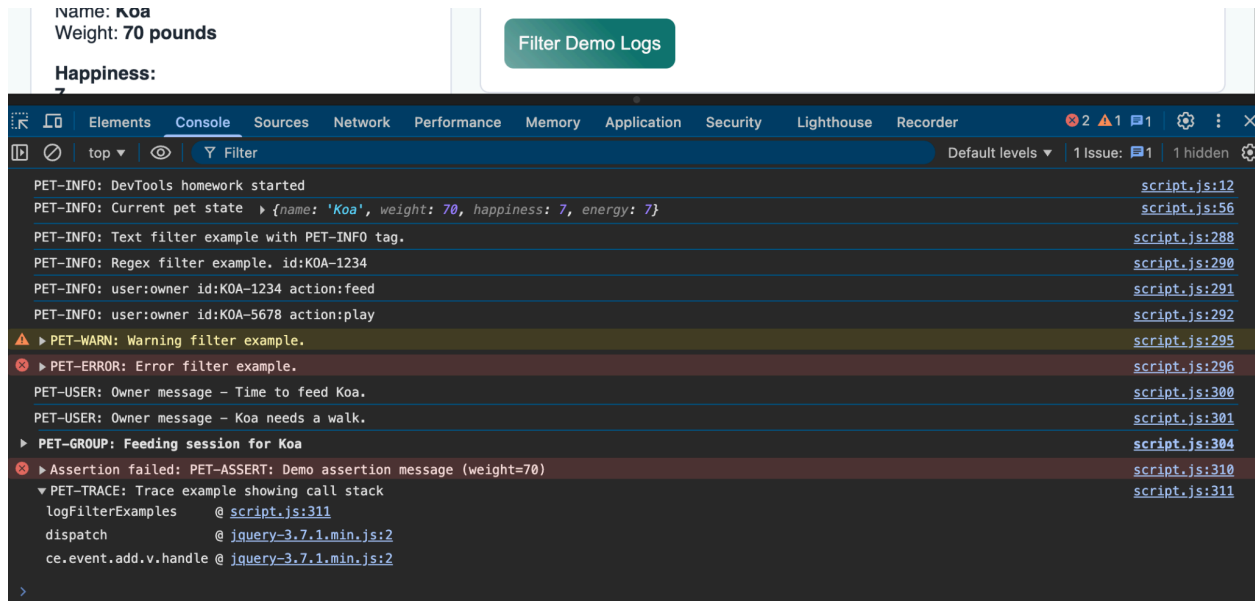


4) Filter Messages

The demo logs are designed to support several DevTools filtering methods:

- Text filtering with tags like ``PET-INFO``, ``PET-WARN``, ``PET-ERROR``, ``PET-USER``, and ``PET-GROUP``.

- Regex filtering with IDs such as `id:KOA-1234`, using a pattern like `/KOA-\d{4}/`.
- Source filtering by searching for `SOURCE-DEMO` or by selecting messages from `script.js`.
- Level filtering with the DevTools level selector: Verbose, Info, Warnings, or Errors.



5) Reproduce a Bug + Sources UI + Apply Fix

The bug occurred because input values retrieved using `$('#number-one').val()` and `$('#number-two').val()` were treated as strings. When added using `+`, this resulted in string concatenation (e.g., `"5" + "1" → "51"`) instead of numeric addition.

Using the browser's debugging tools, a breakpoint was set in `runBuggyAddition()` to inspect variable types. The Scope panel and a Watch expression (`typeof sum`) confirmed that the operands were strings, causing the incorrect behavior.

The issue was fixed by converting the inputs to numbers before performing the addition using `Number()`. This ensured correct numerical results (e.g., `5 + 1 = 6`).

Sources Debugging Lab

This section has a small bug on purpose so I can use the Sources tab, breakpoints, Scope, Watch, and the Console to fix it.

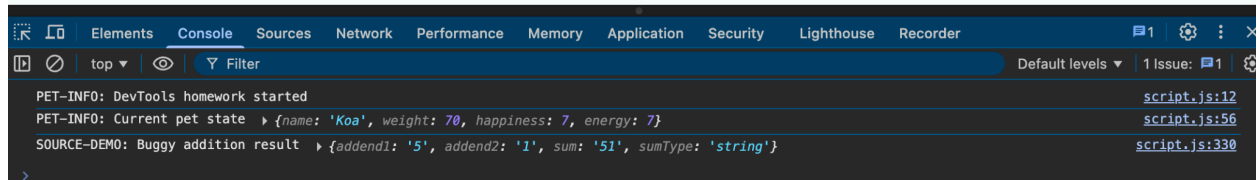
Number 1

Number 2

Buggy Add

Fixed Add

5 + 1 = 51 (bug reproduced)



Sources Debugging Lab

This section has a small bug on purpose so I can use the Sources tab, breakpoints, Scope, Watch, and the Console to fix it.

Number 1

Number 2

Buggy Add

Fixed Add

5 + 1 = 6 (fixed)

